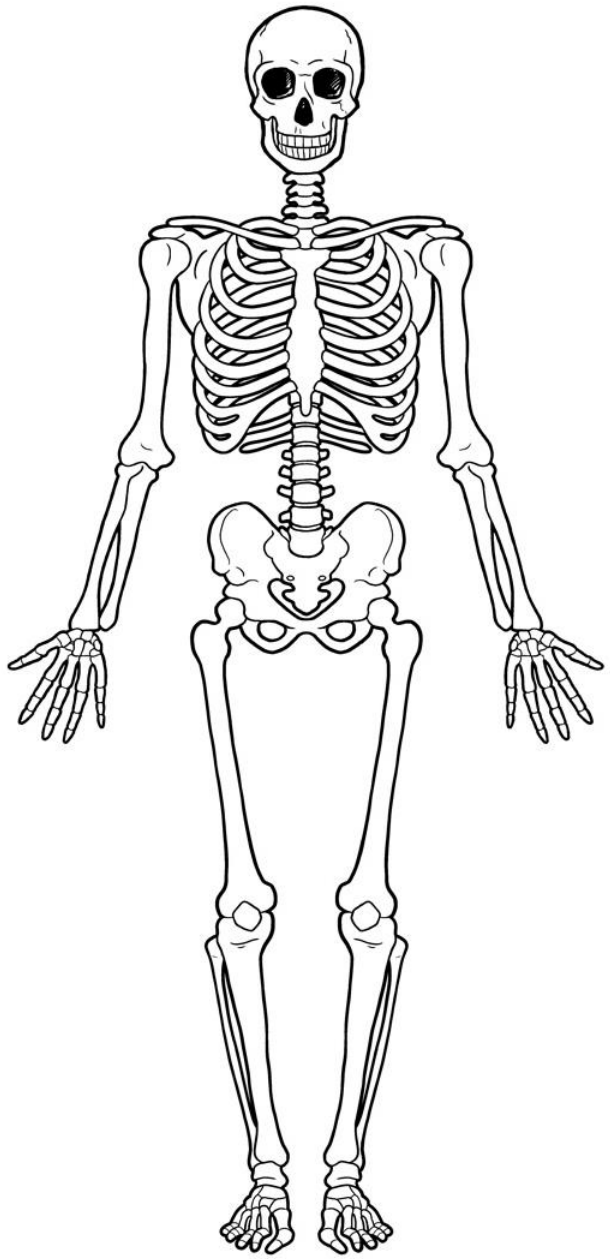
Principles of Human Physiology

Skeletal System



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Functions of Skeletal System

- The human skeletal system consists of bones, cartilage, ligaments and tendons and accounts about 20% of body weight.
- The living bones use oxygen and give off waste products in metabolism.
- Bones provide a rigid frame work (skeleton) that support and protect the soft organs of the body.
- The skeleton supports the body against the pull of gravity. The large bones of lower limbs support the trunk when standing.
- The fused bones of cranium surround the brain, Vertebrae surround and protect the spinal cord and the rib cage protects the heart and lungs.

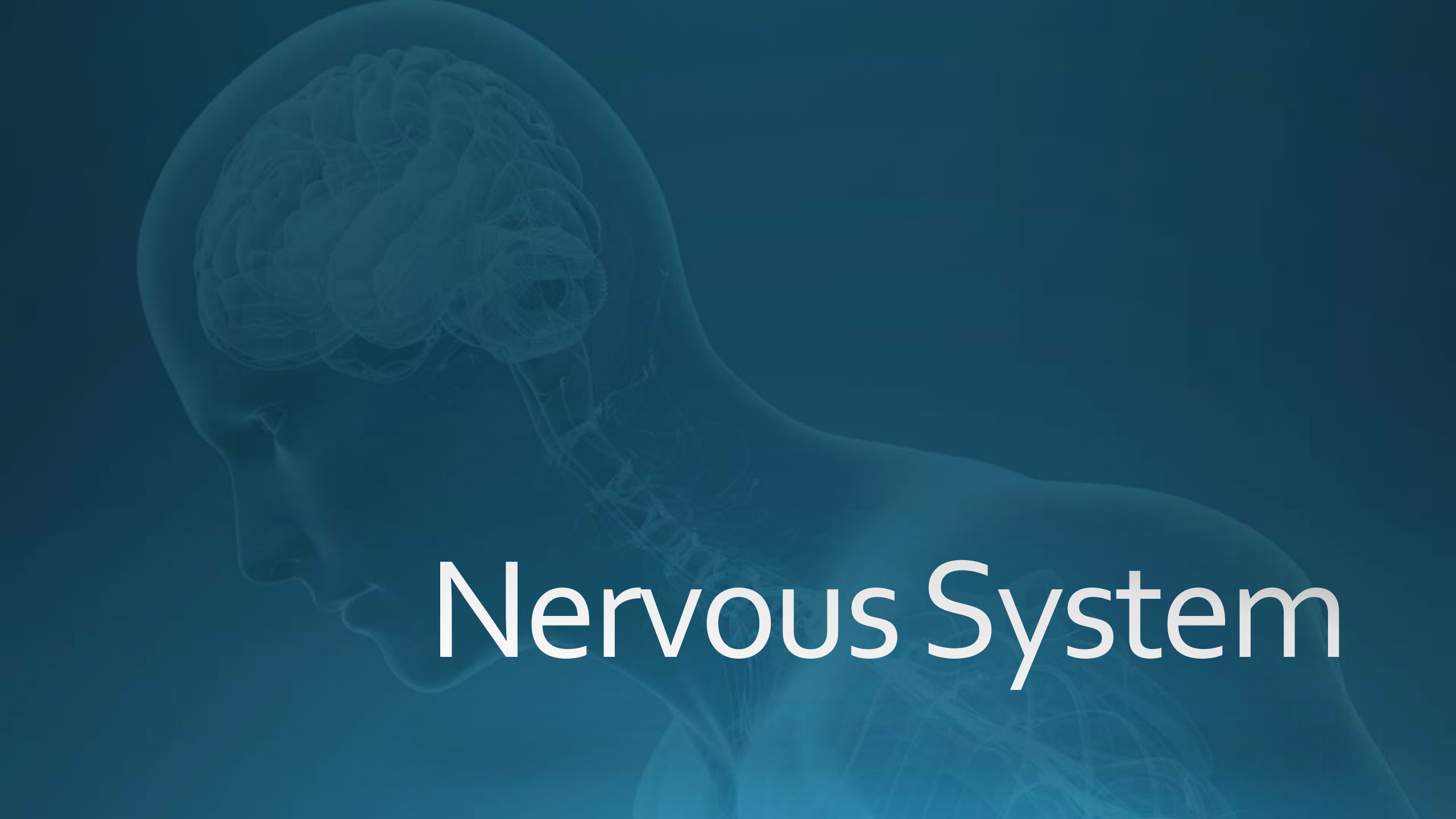
Functions of Skeletal System

- Bones with muscles work together to produce body movement.
- Bones contain more calcium than any other organ. When blood calcium levels decrease below normal, Calcium is released from the bones so that there will be adequate supply for metabolism.
- Hematopoiesis, the formation of blood cells, mostly take place in the red marrow of the bones.

Muscular System

Functions of Muscular System

- The Muscular tissue is composed of muscle fibers. Their predominant function is contractibility.
- Muscles, where attached to bones and blood vessels are responsible for movement.
- The integrated action of joints, bones and skeletal muscles produces obvious movement such as walking and running.
- Skeletal muscle also produce more subtle movement that result in various facial expression, eye movement and respiration.
- Nearly 85% of the heat produced in the body is the result of muscle contraction.



Nervous System

Functions of Nervous System

The Nervous system is the major controlling, regulatory and communicating system in the body.

The center of all mental activity including though, learning and memory.

Together with the endocrine system, the nervous system is responsible for regulating and maintaining homeostasis.

The various activities of the Nervous system can be grouped together as 3 general overlapping functions:

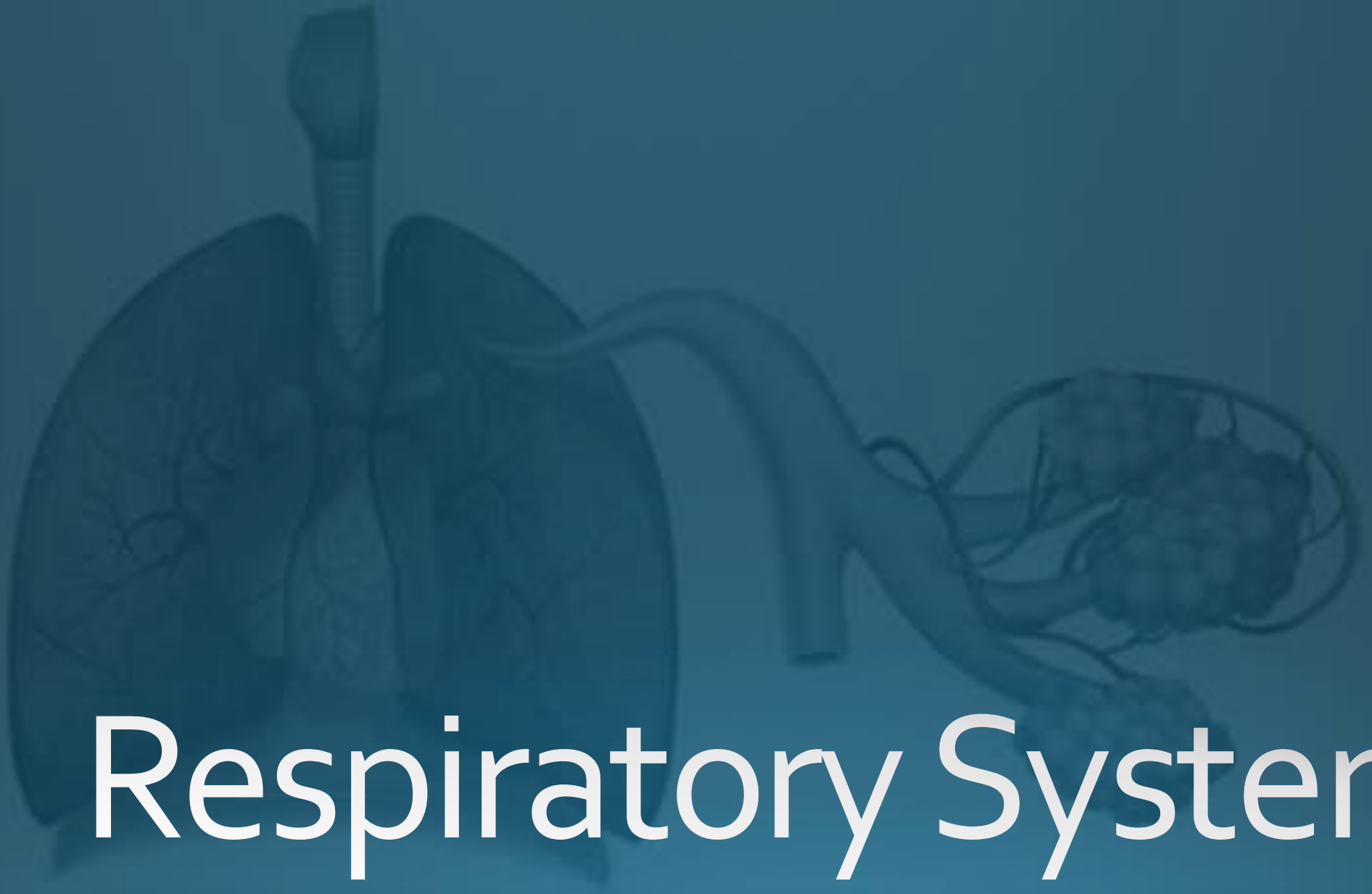
Sensory.	Integrative.	Motor.

Functions of Nervous System

<u>Sensory:</u> Millions of sensory receptors detect changes (Stimuli), which occur inside or outside the body. (Temp., light, PH)

<u>Integrative</u>: Sensory input is converted into electrical signals called nerve impulses that are transmitted to the brain. Decisions are made each moment based on the sensory input.

<u>Motor</u>: Based on the sensory input and integration, the NS responds by sending signals to muscles, causing them to contract or to glands causing them to produce secretions.



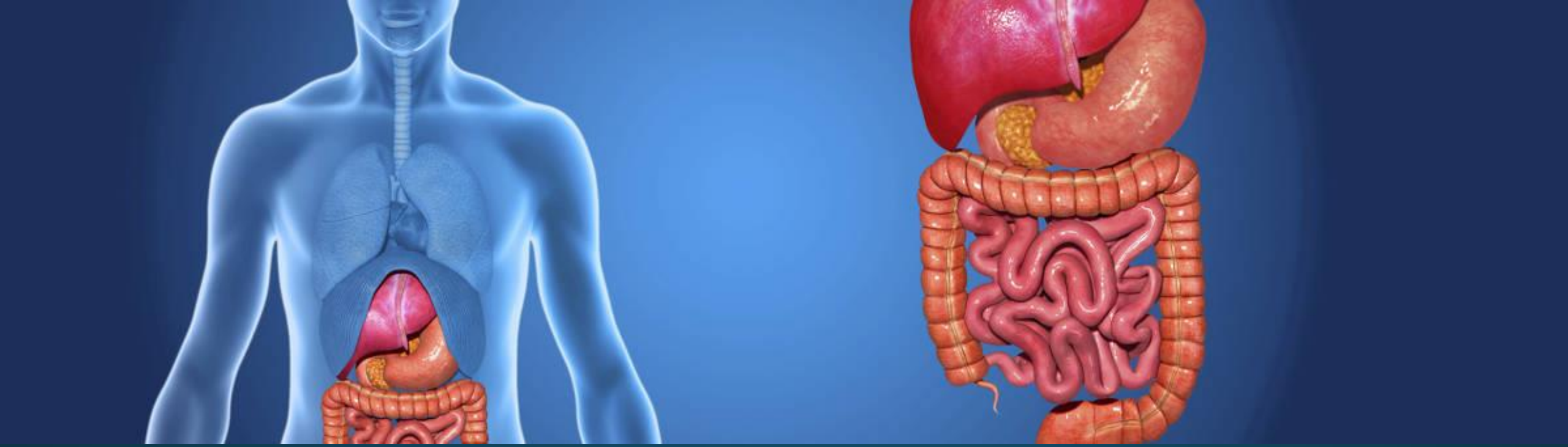
Respiratory System

Functions of Respiratory System

Respiration is the sequence of vents that results in the exchange of O_2 and CO_2 between the atmosphere and body cells. After this, exchanges of gases between lungs and blood (External respiration). Then, from blood to tissues (Internal respiration).

The respiratory system works with the circulatory system to provide O_2 and to remove the waste products of metabolism.

It helps to regulate PH of the blood.



Digestive System

Functions of Digestive System

- Includes the digestive tract and its accessory organs, which process food into molecules that can be absorbed and utilized by the cells of the body.
- Food undergoes 3 types of processes in the body:
  - Digestion.
  - Absorption.
  - Assimilation.
  - Elimination.
- Ingestion: the 1st activity of digestive system is to take in food through the mouth.

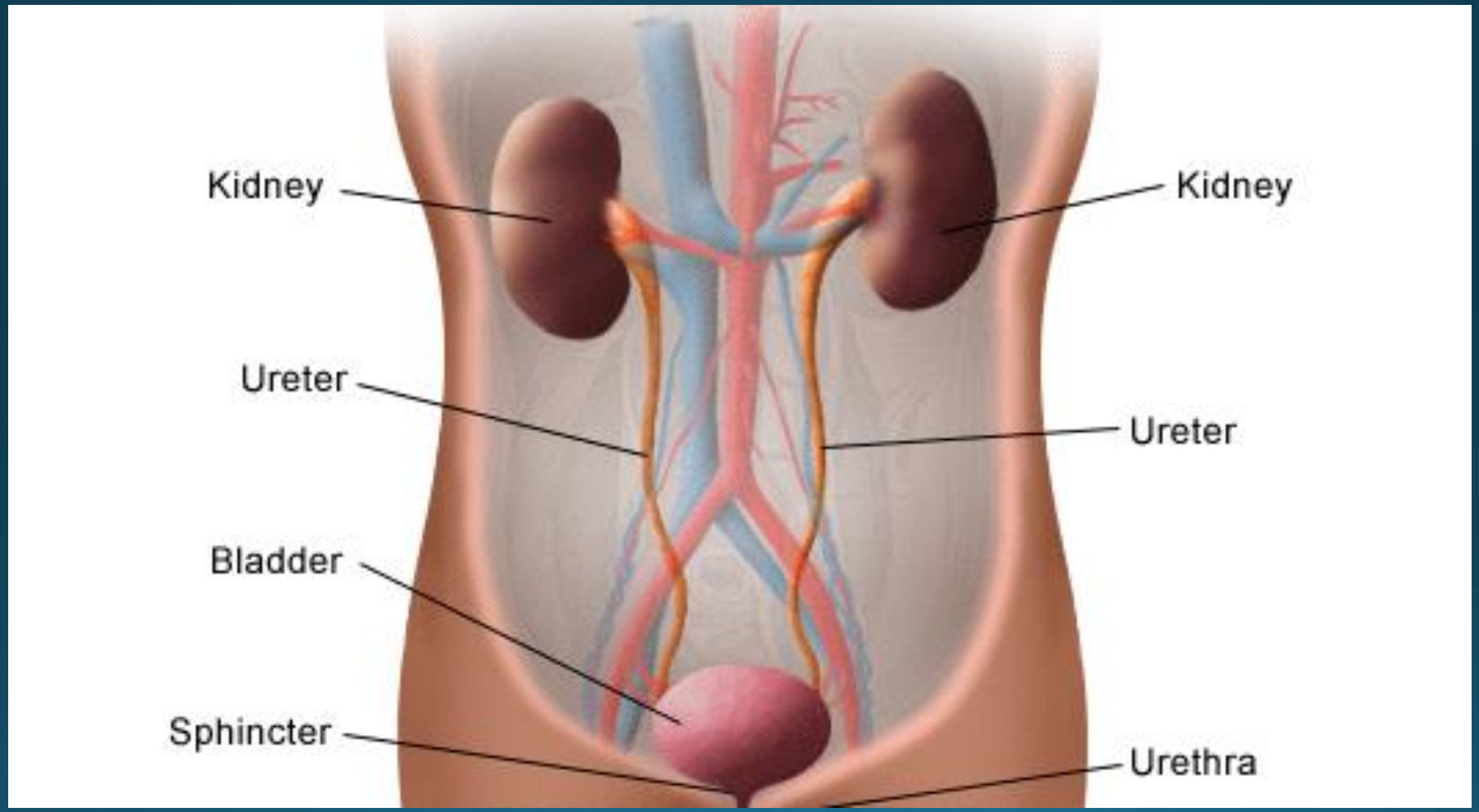
Functions of Digestive System

- <u>Mechanical Digestion</u>: the large pieces of food that are ingested have to broken into smaller particles. Begins by chewing at mouth and continues with churning and mixing in stomach.
- <u>Chemical Digestion</u>: through a process called hydrolysis, uses water and digestive enzymes to break down the complex molecules.
- <u>Movement</u>: is the swallowing the food from mouth to pharynx, then into esophagus. These are rhythmic waves of contraction that move the food particles.

Functions of Digestive System

- **Absorption: simple molecules that result from chemical digestion pass through cell membranes of the lining in the small intestine into the blood or lymph capillaries.**
- **Elimination: food molecules that can't be digested or absorbed need to be eliminated from the body. The removal of indigestible wastes through the anus is elimination.**

Urinary System



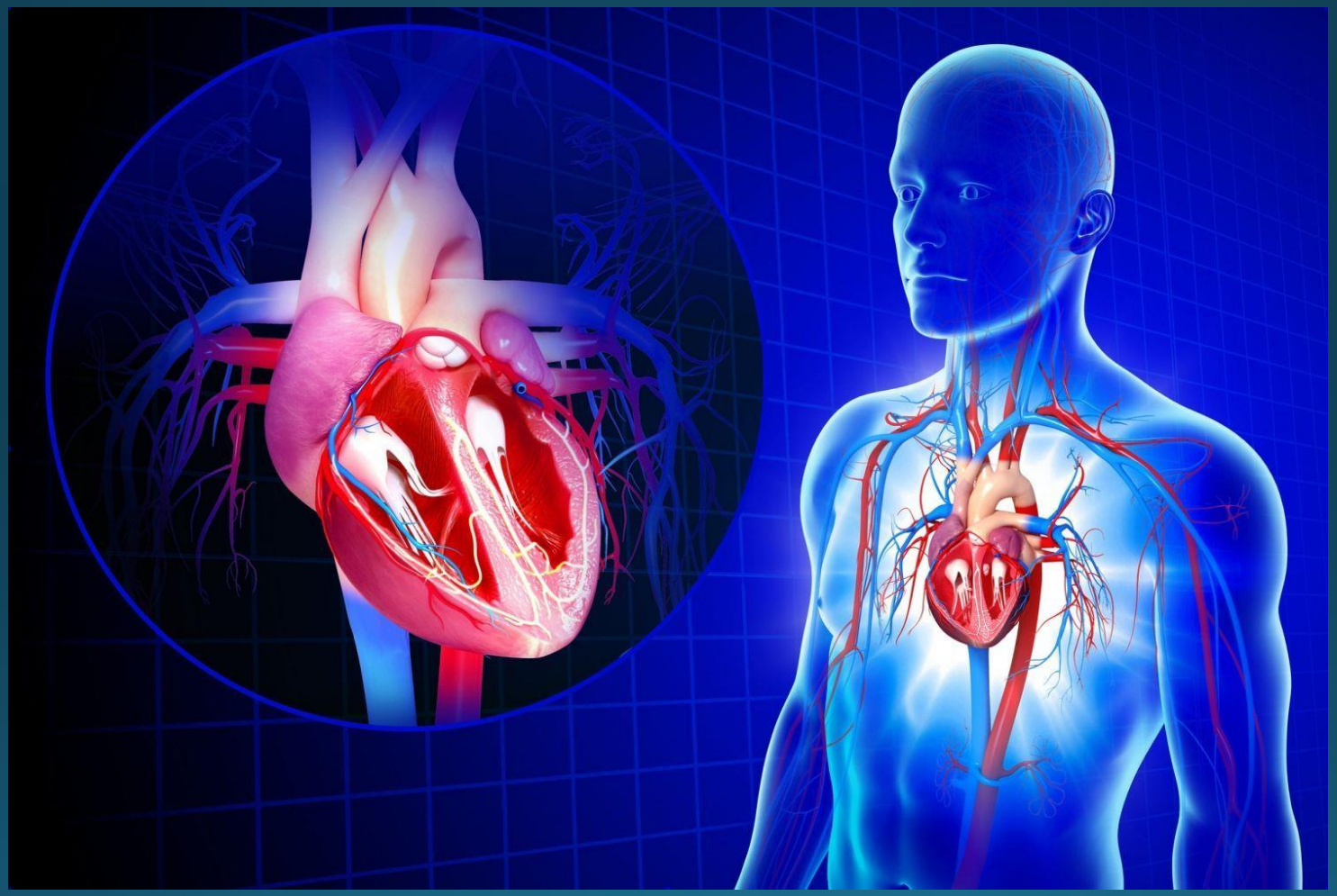
Functions of Urinary System

- The main function of the urinary system is to maintain the volume and composition of body fluids within normal limits by regulating the amount of water that is excreted in the urine.
- Regulating the concentration of various electrolytes in the body fluids and maintaining normal PH of the blood.
- Control red blood cell production by secreting the erythropoietin hormone.
- Plays a role in maintaining normal blood pressure by secreting renin enzyme.

Functions of Reproductive System

- The main function of the reproductive system is to ensure survival of the species.
  - Produce egg and sperm cells.
  - Transport and sustain these cells.
  - Nurture the developing offspring.
  - Produce hormones.

Cardiovascular System



Functions of Cardiovascular System

- It consists of the heart and a closed system of vessels called arteries, veins and capillaries.
- Survival of the developing embryos depending on the circulation of blood to maintain homeostasis.
- The vital role is maintaining homeostasis depends on continuous and controlled movement of blood to every cell in the body.
- Blood transport nutrients and other essential materials to cells.
- Blood remove waste products from the body.